TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Words: at least ≥ · at most ≤ · more than > · fewer than <.
Solve like an equation: undo + and −, then × and ÷.
Flip the sign when you multiply or divide both sides by a negative number.
Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. What value of x makes $2x + 5 = 17$ true?

A. 11

B. 8.5

C. 6

D. 12

2. Which describes the graph of $x \geq 2$ on a number line?

F. open circle at 2, shaded right

G. closed circle at 2, shaded left

H. open circle at 2, shaded left

J. closed circle at 2, shaded right

3. Is $x = -5$ a solution of $-x > 4$?

A. Yes, because $5 > 4$

B. No, because $-5 < 4$

C. Yes, because $-5 > 4$

D. No, because $5 < 4$

4. Chloe buys 2 muffins for $1.50 each and some juices for $2.25 each. Which equation gives the total cost, y, for x juices?

F. $y = 1.50x + 4.50$

G. $y = 2.25x + 3$

H. $y = 3x + 2.25$

J. $y = 3.75x$